

1. **Confident in your confidence intervals?** In lecture we formed a randomization-based confidence interval for the difference in mean response between the $X = 11$ group and the $X = 73$ group in the anchoring experiment. Your goal for this exercise is to use simulation to check the coverage of your interval and evaluate the consequences of using the constant effect assumption when it is violated.

This simulation requires knowing the value of the true parameter, ATE , so you'll be playing this question as an all knowing god with all potential outcomes at your fingertips. You can find two full schedules of potential outcomes at the following links.

```
godlikeA <- read.csv("https://stat158.berkeley.edu/spring-2026/data/anchoring/
godlike_schedule_A.csv")
godlikeB <- read.csv("https://stat158.berkeley.edu/spring-2026/data/anchoring/
godlike_schedule_B.csv")
```

Schedule A represents a world with constant effects: the individual treatment effect for each unit is the same. Schedule B represents a world with variable effects: each unit has a different individual treatment effect.

- a. For each god-like schedule A and B, calculate the true ATE parameters.
- b. For god-like schedule A, use simulation to create many original experiments and then for each construct a randomization-based CI with a constant effect assumption and evaluate whether it contains the true parameters. Select a $1 - \alpha$ confidence level to use that is between 60% and 90%. For A and B separately, report the proportion of confidence intervals that contain the parameter.
- c. Repeat b. but use god-like schedule B.
- d. Did your randomization-based CIs with the constant effect assumption from the previous two questions succeed in capturing the parameter with probability $1 - \alpha$? If not, provide some intuition for why not.

This is a challenging simulation to set up correctly. If you get stuck please consult the hints¹²³.

¹The idea is to recreate the random process that leads to the create of the random confidence interval many times and check the proportion of those intervals that contain the parameter. In randomization-based inference, the only randomness comes from the assignments of units to treatment and control at the start of the original experiment, so that's where each one of your simulations begins.

²For each of settings A and B above, you will want to write a function. One approach is to write a function that:

1. Randomly assigns each unit to one of the two groups.
2. Sets the potential outcome that is *not* observed to be NA. At this stage, you have the analog of an original experimental data set.
3. Calculates $\widehat{ATE}$.
4. Uses $\widehat{ATE}$ to impute the missing potential outcomes.
5. Using this filled-in schedule, form a randomization-based confidence interval.
6. Check whether the interval contains the true parameter and return that logical value.

Then you can replicate this function many times and count the proportion of TRUEs to estimate the coverage probability. Note that if you use the code from the slides, you will be pivoting the data back and forth between schedule-format (with a column for each potential outcome) and dataframe-format (with all of the observed outcomes in a single column).

³This code may take awhile to run - you're simulating many randomization-based CIs - so start with small numbers of replicates while you're still testing your code. You can also take advantage of your computers ability to run multiple R sessions at the same time using the `future_replicate()` function inside the `future.apply` package.

More questions coming soon!